4. Draw a neat labelled diagram and explain the salient features of Watson-Crick Structure of DNA.

5. Outline the experiment carried out by Griffith to identify the genetic material.

6. With a neat illustration, explain the structure of t-RNA.

7. Give the differences between DNA and RNA.

Activity

The students may be asked to prepare a three dimensional structure of DNA or to prepare a chart explaining the structure of DNA.

SUMMARY

Nucleic acids are the molecular repositories of genetic inheritance. Every macro molecules of the cell is the product of the information that has been stored in the nucleotide sequences of genes. Nucleic acids are long threadlike polymers made up of monomers called nucleotides held together by phospho diester bridges. Nucleotides have three components namely base, sugar and a phosphate group.

The bases are either monocyclic pyrimidines (cytosine, uracil and thymine) or the bicyclic purines (adenine and guanine). There are two types of sugars are present in nucleic acid. They are ribose and deoxyribose. When the sugar is ribose it is called ribonucleic acid (RNA) and if the sugar is deoxy ribose it is called Deoxyribonucleic acid (DNA). The phosphate groups are used to connect the nucleotides with each other. Depending upon the number of phosphate groups they can be classified as monophosphates, diphosphates and triphosphates.

DNA is the genetic material of all living organisms except RNA viruses like HIV. The B form DNA, also known as the Watson-crick DNA, is the most stable and prevalent form of DNA. The two antiparellel strandes of DNA are complementary to each others. The diemeter of DNA is 2nm and helical length is 3.4nm. It has ~10.5basepairs per turn of the helix. It has three different has three different forms namely A-DNA, B-DNA and Z-DNA. There are three major types of RNA are known. They are messenger RNA (mRNA), transfer RNA (tRNA) and ribosomal RNA (rRNA).

They differ from each other by size, shape, formation and stability. The mRNA is a direct carrier of genetic information from nucleic acid to cytoplasm. tRNA is the carrier of amino acids to the site of protein synthesis. rRNA are required or the formation of ribosomes.

unit 7.indd 194 21-05-2018 15:11:19